

FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING (25CS1201E)
2025-26, Term-III
Project Problem Statement Submission Form

Sec No-	Team No-	Project Title-
Team Lead Roll Number		2520030128
Team Lead Name		RC Sharan Sai
Team Member Roll Number		2520030075
Team Member Name		Thyarla Abhinav

1. Abstract

(Summarize the project idea, problem background, and proposed solution. Not more than 250 words.)

The **Pharmacy Inventory Management System** is a web-based solution designed to efficiently manage drug stock, monitor expiry dates, and automate pharmacy operations. Built using modern front-end technologies and component-based architecture, the system ensures a scalable, responsive, and user-friendly interface. It utilizes JavaScript (ES6+) concepts such as modular design, asynchronous data handling, and efficient state management to provide real-time updates and smooth user interactions. Key features include inventory tracking, supplier management, reorder alerts, and API-based data integration.

The system also focuses on performance optimization, usability, and reliability, making it suitable for real-world deployment. Overall, it enhances accuracy, reduces manual effort, and improves the efficiency of pharmacy management processes.

2. Module Mapping

(Provide details of how modules are divided among team members.)

Module Name	Assigned Roll Number(s)	Signature
Authentication & admin dashboard	2520030128	
Inventory & stock management	2520030075	
Supplier & recorder management	2520030128	
Expiry alerts, reports & notifications	2520030075	

3. Module Details

(Explain the responsibility of each student concerning their assigned module.)

RC Sharan Sai – Roll No: 2520030128

- Develop the login/authentication system and admin dashboard.
- Implement supplier order management and reorder request features.
- Integrate REST APIs for handling medicine and supplier data.

- Design responsive UI components using React.js and Tailwind CSS.
- Manage routing and navigation using React Router.

Thyarla Abhinav – Roll No: 2520030075

- Develop inventory and stock management modules.
- Implement expiry date tracking and alert notifications.
- Create dispensing records and barcode lookup functionality.
- Build analytics charts and export features for reports.
- Handle state management and local storage integration.

4. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

HTML5

CSS3

JavaScript (ES6+)

React.js

Node.js

Express.js

npm

Axios

React Router

Context API / Redux

Tailwind CSS / Bootstrap

REST APIs

LocalStorage

Postman (API Testing)

MySQL (Optional)

VS Code

Git & GitHub

Guide Name and Signature (remarks if any) _____